

HEAVEN'S LIGHT IS OUR GUIDE



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DEPARTMENT OF MECHATRONICS ENGINEERING**

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An Automated Greenhouse System with Efficient Climate, Irrigation and Lighting Control with Fault Recovery Approach

Introduction

Automation refers to the use of control systems, such as Programmable Logic Controllers (PLCs), sensors, and actuators, to perform tasks with minimal human intervention [1]. It has become one of the central pillars of modern industry, enabling processes to run with greater precision, consistency, and efficiency than manual operation allows. The theoretical foundation of automation lies in control theory, where feedback loops and logical decision-making are used to regulate the behaviour of a system in response to changing conditions [2][3].

The importance of automation has grown substantially with the emergence of Industry 4.0, a paradigm that integrates cyber-physical systems, the Internet of Things (IoT), and real-time data exchange into industrial and agricultural processes [4]. Within this paradigm, automation is no longer limited to isolated machines performing repetitive tasks; instead, interconnected sensors and controllers communicate continuously, allowing systems to adapt dynamically to their environment. This shift is particularly significant in sectors such as agriculture, where precision automation enables optimized resource utilization, improved yield consistency, and reduced dependency on manual labour [5][6].

Globally, the state of automation varies widely across industries and regions. Developed economies have widely adopted advanced automation technologies, including PLC-based industrial control, SCADA systems, and IoT-integrated smart agriculture, as demonstrated in large-scale precision agriculture deployments such as the smart greenhouse initiatives implemented in Qatar [7]. In contrast, the adoption of automation in developing countries, including Bangladesh, remains comparatively limited. Several challenges hinder widespread implementation, including the high initial cost of automation hardware, a shortage of technically skilled personnel capable of designing and maintaining such systems, inconsistent electricity supply in rural areas, and limited awareness among traditional farmers and small-scale industry operators regarding the long-term benefits of automation [8].

Despite these challenges, several practical solutions can accelerate the adoption of automation in Bangladesh. Government and academic institutions can play a pivotal role by incorporating hands-on automation and PLC training into engineering curricula, thereby producing a workforce capable of designing, deploying, and maintaining such systems locally. Additionally, the development of low-cost, modular automation solutions tailored to the scale of local farms and industries can

reduce the financial barrier to entry. Public-private partnerships aimed at rural electrification and subsidized automation equipment could further support this transition, enabling Bangladesh to gradually bridge the gap between traditional manual practices and modern automated systems [4].

This report presents the design and simulation of an automated greenhouse control system as a practical application of PLC-based automation, addressing climate control, precision irrigation, fertilizer dosing, grow lighting, and fault detection within a single integrated ladder logic framework.

Required Kits:

- i. LOGO!Soft Comfort V7.0

I/O Mapping Table

Table 01: Analog Inputs

Address	Label	Full Name	Function	Stage
AI1	T_S_A	Temperature Sensor Zone A	Measures the temp of Zone A	Stage 01
AI2	H_S_A	Humidity Sensor Zone A	Measures the humidity of Zone A	Stage 01
AI3	T_S_B	Temperature Sensor Zone B	Measures the temp of Zone B	Stage 01
AI4	H_S_B	Humidity Sensor Zone B	Measures the humidity of Zone B	Stage 01
AI5	P_S	Pressure Sensor	Measure the Pressure of Irrigation Pump Line	Stage 02
AI6	L_L_S	Low Level Sensor	Measures Level of Fertilizer Tank	Stage 04
AI7	A_L_A	Ambient Light Zone A	Measures the Ambiance of Zone A	Stage 05
AI8	A_L_B	Ambient Light Zone B	Measures the Ambiance of Zone B	Stage 05

Table 02: Digital Inputs

Address	Label	Full Name	Function	Stage
I1	W_S_S	Wind Speed Sensor	Detects Wind	Stage 01
I2	R_S	Rain Sensor	Detects Rain	Stage 01
I3	I.Z.A	Irrigation Zone A	Detects Moisture in Zone A	Stage 02
I4	I.Z.B	Irrigation Zone B	Detects Moisture in Zone B	Stage 02
I5	I.Z.C	Irrigation Zone C	Detects Moisture in Zone C	Stage 02
I6	S.B	Set Button	Arms the pressure fault latch	Stage 02
I7	R.B	Reset Button	Reset the Pressure Fault Alarm	Stage 02
I8	S.B	Set Button	Arms the low-level fault latch	Stage 04
I9	R.B	Reset Button	Reset the Low-Level Sensor Fault Alarm	Stage 04

Table 03: Analog Comparators

Address	Label	Full Name	Function	Stage
SF001	T_C_A	Temperature Comparator Zone A	Compares the Temperature of Zone A with Threshold Value	Stage 01
SF002	H_C_A	Humidity Comparator Zone A	Compares the Humidity of Zone A with Threshold Value	Stage 01
SF004	T_C_B	Temperature Comparator Zone B	Compares the Temperature of Zone B with Threshold Value	Stage 01
SF005	H_C_B	Humidity Comparator Zone B	Compares the Humidity of Zone B with Threshold Value	Stage 01
SF007	L.P.C	Low Pressure Comparator	Compares the Pressure with Low Pressure Value	Stage 02
SF015	H.P.C	High Pressure Comparator	Compares the Pressure with High Pressure Value	Stage 02
SF009	L.L.S.C	Low-Level Sensor Comparator	Compares the Fertilizer Tank Level with Threshold Value	Stage 04
SF010	A.C.A	Ambiance Comparator Zone A	Compares the Ambiance of Zone A with Threshold Value	Stage 05
SF011	A.C.B	Ambiance Comparator Zone B	Compares the Ambiance of Zone B with Threshold Value	Stage 05

Table 04: SR Latch (Set-Reset Relay)

Address	Label	Full Name	Function	Stage
SF012	R_O_P	Reset Operation for Pressure	Resets the Pressure Fault Alarm	Stage 07
SF014	R_O_L_L	Reset Operation for Low-Level Tank	Resets the Low-Level Tank Fault Alarm	Stage 07

Table 05: Time Delay

Address	Label	Full Name	Function	Stage
T003	T_D_A	Time Duration for Fan A	Allows to run the fan in Zone A for a time duration	Stage 01
T006	T_D_B	Time Duration for Fan B	Allows to run the fan in Zone A for a time duration	Stage 01

Table 06: Outputs

Address	Label	Full Name	Function	Stage
Q1	C_F_A	Circulation Fan Zone A	Improving Air Exchange in Zone A	Stage 01
Q2	V_M_A	Vent Motor Zone A	Improving Air Exchange in Zone A	Stage 01
Q3	C_F_B	Circulation Fan Zone B	Improving Air Exchange in Zone B	Stage 01
Q4	V_M_B	Vent Motor Zone B	Improving Air Exchange in Zone B	Stage 01
Q5	S_V_A	Solenoid Valve Zone A	Controlling the Irrigation Zone A	Stage 02
Q6	S_V_B	Solenoid Valve Zone B	Controlling the Irrigation Zone B	Stage 02
Q7	S_V_C	Solenoid Valve Zone C	Controlling the Irrigation Zone C	Stage 02
Q8	I_P	Irrigation Pump	Moisturizing Dry Soil	Stage 02
Q9	F_A_P	Fault Alarm for Pressure	Indicating the Fault in the Pressure in Irrigation Pump Line	Stage 07
Q10	F_D_V	Fertilizer Dosing Valve	Dosing Fertilizer by Valve	Stage 04

Address	Label	Full Name	Function	Stage
Q11	F_A_L_L	Fault Alarm for Low-Level in Tank	Indicating the Fault in the Tank	Stage 07
Q12	L_A	Light Zone A	Light for Zone A in Insufficient Natural Light	Stage 05
Q13	L_B	Light Zone B	Light for Zone B in Insufficient Natural Light	Stage 05

Sequential Logic

The sequential logic for all the stages have been described step-by-step in the below section.

Stage 01

Stage 01 describes the Climate Humidity Control of the whole automation system. Here the parameters are: temperature, humidity, wind ar rain. The actuators are: circulation fans and vent motors. If the temperature or the humidity crosses the threshold value, then the circulation fan will be turned on for a timed duration to improve air exchange in each section. But the vent motors depend only on temperature. There are rain sensor for detecting rain, and wind speed sensor for wind. If the rain sensor or the wind speed sensor triggers, then the vent motors will be turned off, no matter whether the temperature crosses the threshold value.

Stage 02

Stage 02 denotes the Precision Zone Irrigation of the automation system. The parameters monitored here are soil moisture and irrigation line pressure. The actuators involved are three solenoid valves (Zone A, B, and C) and one central irrigation pump.

Each irrigation zone operates independently. When the soil moisture sensor of a particular zone detects dry soil, the corresponding solenoid valve opens to allow water flow into that zone. The irrigation pump activates automatically as long as at least one solenoid valve is open, and it stops as soon as all valves are closed.

The irrigation line pressure is continuously monitored by a pressure sensor through two analog comparators; one for low pressure and one for high pressure. If the pressure drops below the lower threshold, it indicates the faulty alarm, and if the pressure rises above the upper threshold, it also indicates the overloading on the pump. Either condition triggers a fault alarm immediately, which latches and stays active until the operator resolves the issue and presses the reset button to restore

normal operation.

Stage 04

Stage 04 specifies the Fertilizer Dosing Control of the automation system. The monitored parameters are the irrigation pump status and the fertilizer tank level, with the fertilizer dosing valve as the only actuator. The dosing valve operates only when the irrigation pump is running, ensuring proper dilution of fertilizer with water inside the irrigation line; if the pump stops, the valve closes immediately. The tank level is monitored by a low-level sensor through an analog comparator, and if the level drops below the threshold, the dosing valve closes and a fault alarm is triggered and latched. The operator must refill the tank and press the reset button to clear the alarm and resume normal dosing operation.

Stage 05

In stage 05, the Grow Lighting Control feature of the automation system is shown. The variables being monitored include the light levels and temperatures in zones, and each zone has only one actuator – grow light. Grow lights will start working automatically if the light level sensor registers that there is not enough sunlight available in a particular zone, thus providing sufficient light for growing crops. However, should the temperature in the zone exceed the threshold, the grow light in that zone will be stopped even though there might not be enough sunlight, since the additional heat from the lights will deteriorate the situation with temperature. Zones work independently of each other and are disconnected from the fault system.

Stage 07

Stage 07 describes the Fault Detection & Recovery of the Automation System. Irrigation line pressure and fertilizer tank level are the monitored parameters in this case, and fault alarms for each fault indicator are provided at output. There is an alarm assigned to each fault situation where if the irrigation line pressure falls below the lower threshold value, it shows the possibility of pipe breakage or leakage; on the other hand, if the irrigation line pressure exceeds the upper threshold limit, it is indicative of pipe blockage, pump overload, or sensor malfunction, as excessive pumping load manifests as abnormally high line pressure. In any case, the alarm latches and remains activated irrespective of the removal of the fault.

Ladder Diagram Implementation

The complete system was implemented in LOGO!Soft Comfort V7 using Ladder Diagram (LAD) programming, with each stage built and verified independently before integration [1].

Stage 1: In Zone A, temperature (AI1) and humidity (AI2) are read through analog comparators SF001 and SF002 respectively to detect high-threshold conditions. The circulation fan (Q1) is driven by an OR combination of the SF001 and SF002 outputs, passed through timer T003 to limit fan run duration, while the vent motor (Q2) is driven solely by the SF001 temperature comparator, gated by a normally closed rain sensor (I2) contact, ensuring the vent closes immediately whenever rain is detected (Fig. 1).

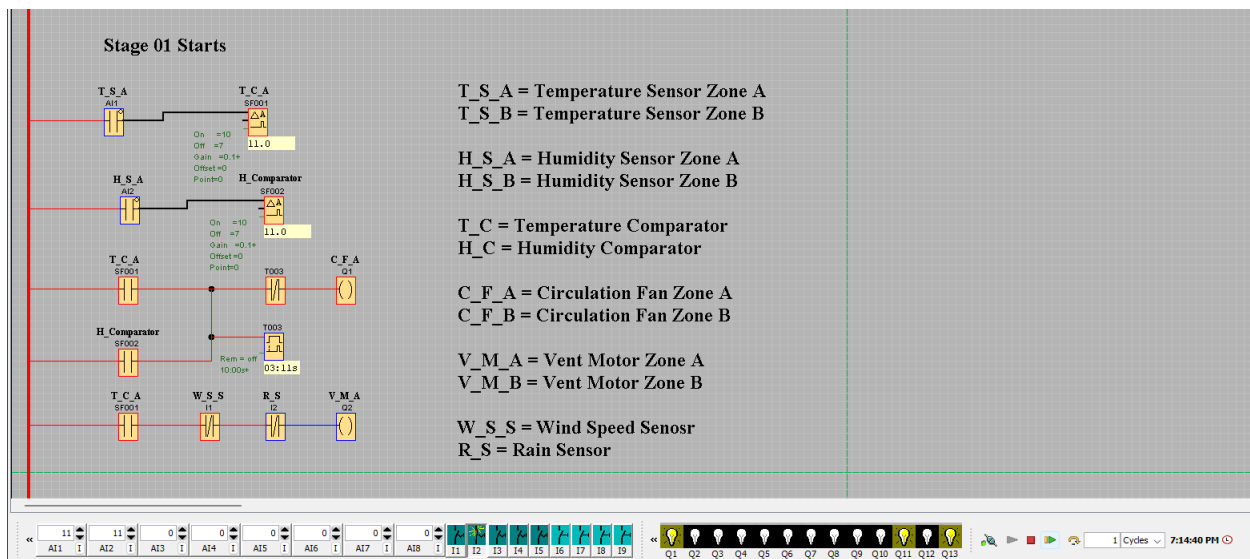


Figure 1: Ladder Diagram for stage 1 of zone A

In Zone B, temperature (AI3) and humidity (AI4) are read through analog comparators SF004 and SF005 respectively, following the same logic structure as Zone A. The circulation fan (Q3) is driven by an OR combination of the SF004 and SF005 outputs, passed through timer T006 to limit fan run duration, while the vent motor (Q4) is driven solely by the SF004 temperature comparator, gated by the same normally closed rain sensor (I2) contact used in Zone A, since a single rain sensor governs vent closure across both zones (Fig. 2).

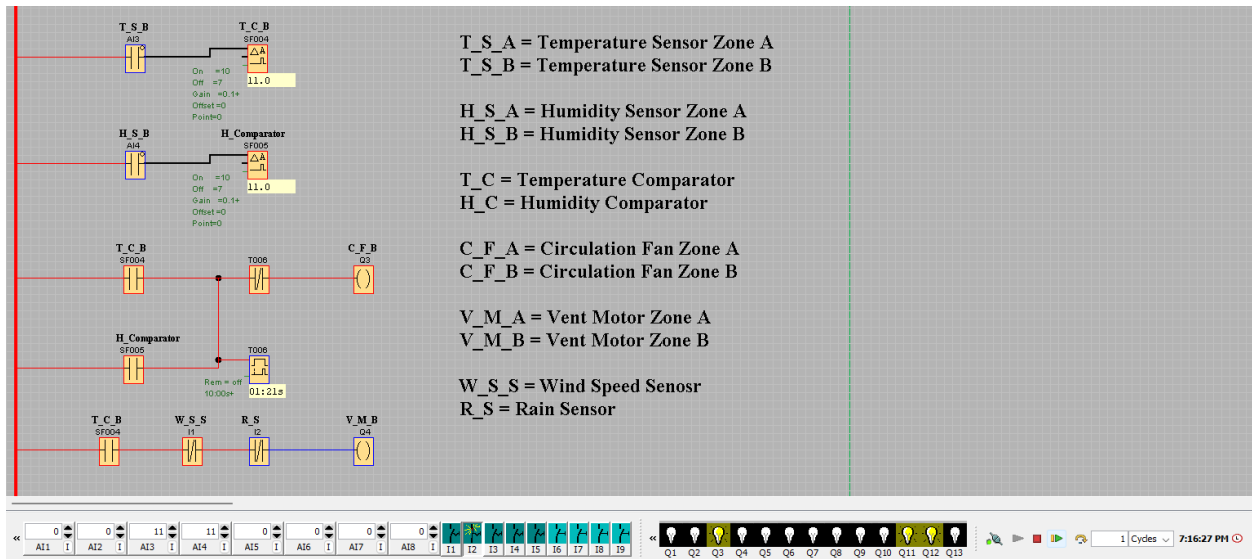


Figure 2: Ladder Diagram for stage 1 of zone B

Stage 2: Each of the three irrigation zones was implemented as an independent rung, where digital soil moisture inputs (I3, I4, I5) directly energize their corresponding solenoid valves (Q5, Q6, Q7). The irrigation pump (Q8) coil is driven by the three valve outputs connected in parallel, so that the pump activates automatically when any zone is irrigating and stops only when all valves are closed (Fig. 3).

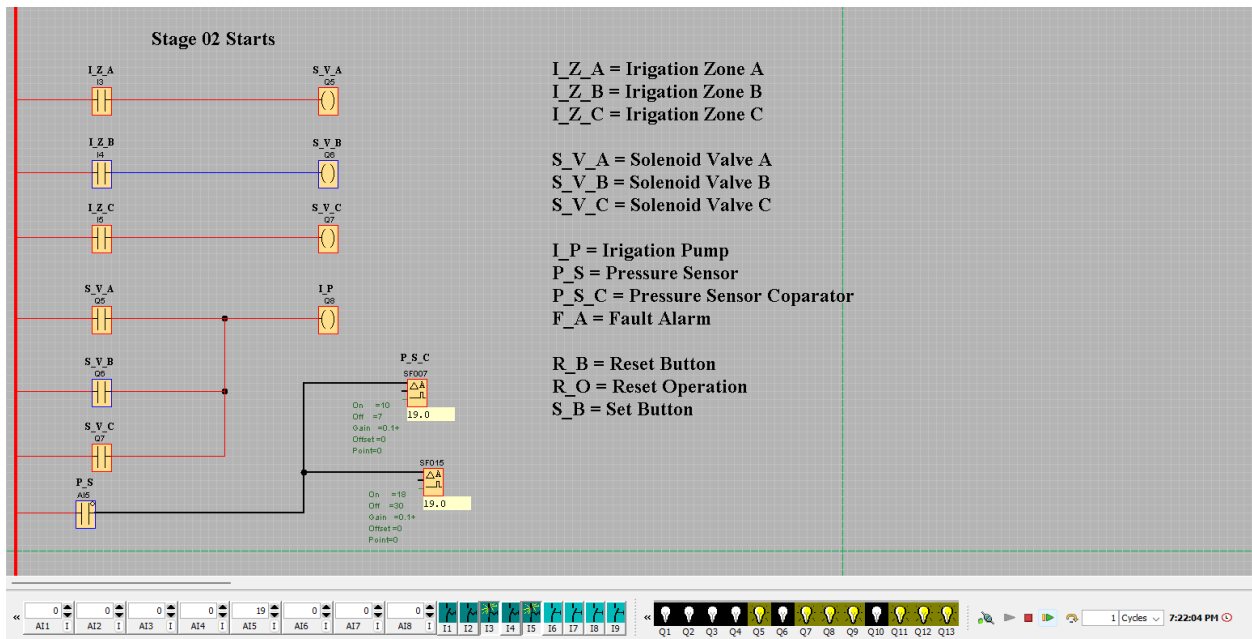


Figure 3: Ladder Diagram for stage 2 of Irrigation Zone

The irrigation line pressure (AI5) is continuously evaluated through two analog comparators — SF007 for low pressure and SF015 for high pressure. A set button (I6) arms the fault latch for this condition, while a dedicated reset button (I7) is provided to clear the pressure fault once the operator has verified normal line pressure (Fig. 4).

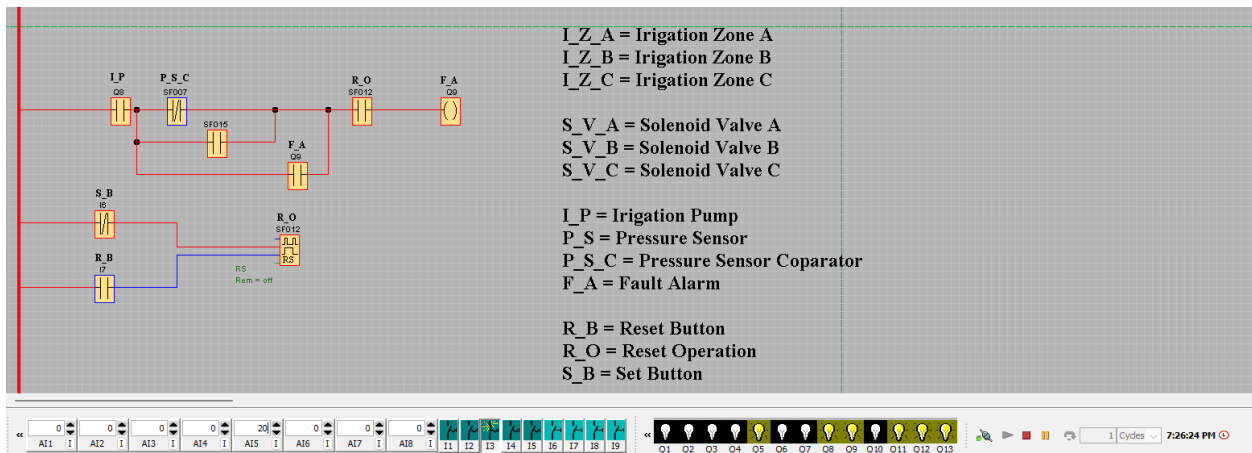


Figure 4: Ladder Diagram for stage 2 of monitoring pressure of irrigation line

Stage 4: The dosing valve (Q10) rung is enabled only when the irrigation pump (Q8) output is active, ensuring fertilizer is always injected into a live water flow, with a timer block controlling the configured dosing interval and injection duration. The fertilizer tank level (AI6) is monitored through analog comparator SF009, whose output inhibits the dosing valve whenever the tank level falls below the set threshold, with a set button (I8) arming the fault latch and a dedicated reset button (I9) provided for clearing the low-level fault condition (Fig. 5).

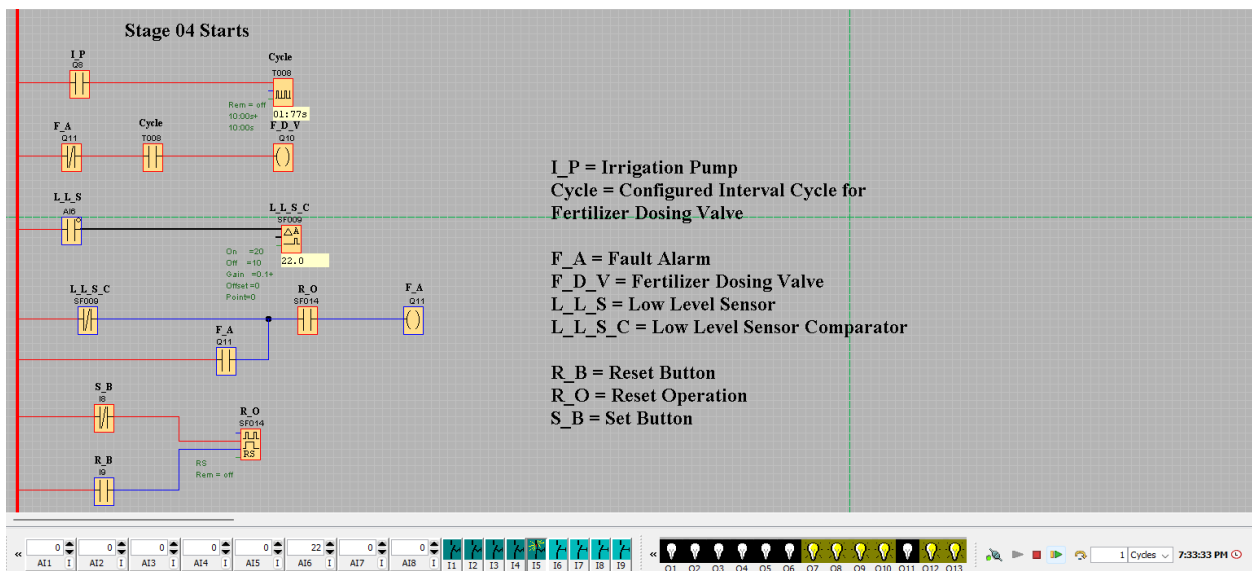


Figure 5: Ladder Diagram for stage 4

Stage 5: Each zone's grow light rung combines two normally closed conditions in series: the ambient light comparator (SF010 for Zone A, SF011 for Zone B), which closes when the natural light level (AI7, AI8) is insufficient, and the corresponding temperature comparator from Stage 1 (SF001 for Zone A, SF004 for Zone B), which opens to inhibit the light (Q12 for Zone A, Q13 for Zone B) whenever the zone temperature is already above its high-limit threshold (Fig. 6).

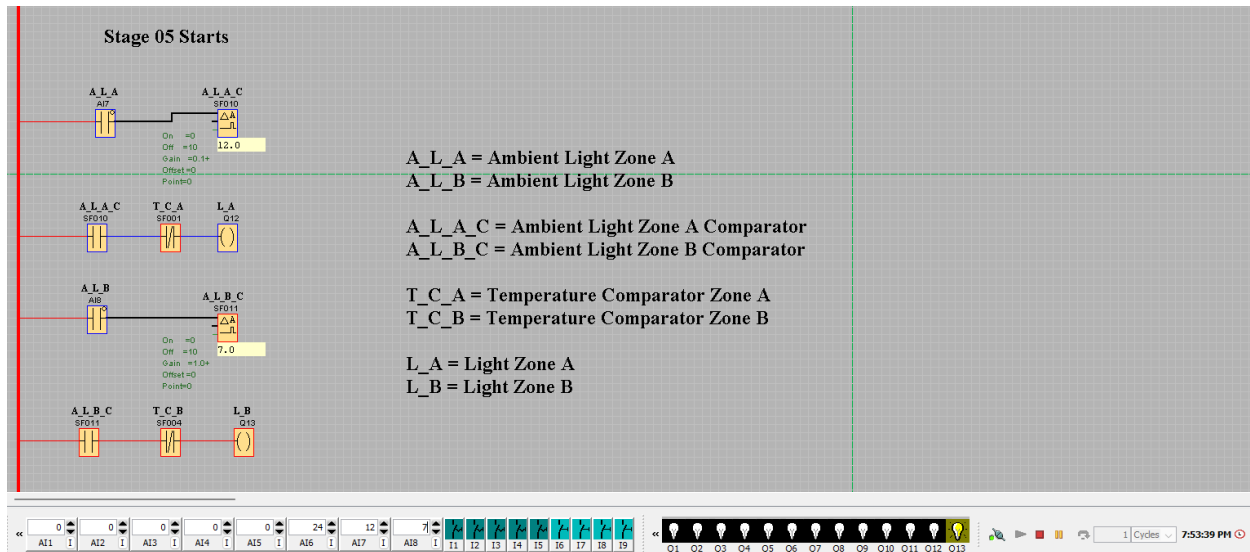


Figure 6: Ladder Diagram for stage 5

Stage 7: Pump overload, pressure loss, and tank-empty conditions were each connected to independent Set-Reset latch circuits, allowing every fault type to trigger and hold its own dedicated alarm output. Each latch is set by its respective fault condition and reset only by a manually operated pushbutton, ensuring the system requires deliberate operator intervention before returning to normal operation after any fault.

The complete LOGO!Soft Comfort V7.0 project file, containing the full ladder diagram implementation for all five stages, is available at:

<https://github.com/shimanto008/greenhouse-plc-automation>.

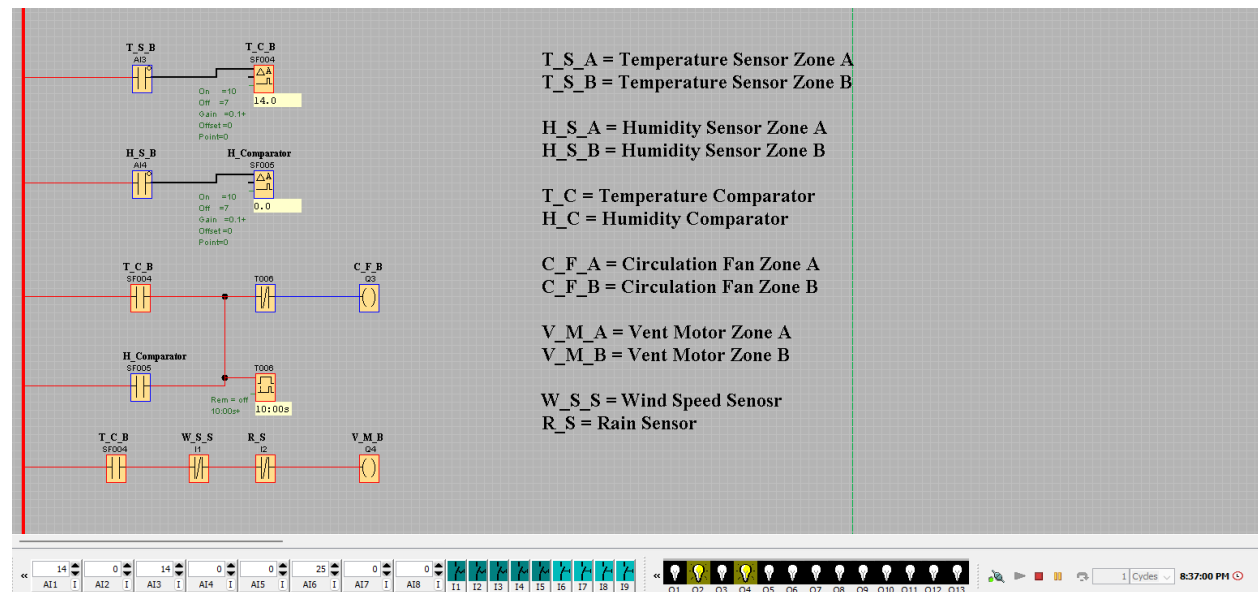
Simulation Results

The completed ladder logic was validated in LOGO!Soft Comfort V7 by running four distinct simulation scenarios, each targeting a specific operating condition of the system.

Normal Operation: With the Zone A temperature (AI1) and humidity (AI2) inputs held within their normal operating ranges, comparators SF001 and SF002 remained inactive, keeping the cir-

[illegible]

Similarly, with the Zone B temperature (AI3) and humidity (AI4) inputs held within their normal operating ranges, comparators SF004 and SF005 remained inactive, keeping the circulation fan (Q3) and vent motor (Q4) both off, confirming identical baseline behaviour for Zone B (Fig. 8).



With all three soil moisture inputs (I3, I4, I5) held in the wet state, the solenoid valves (Q5, Q6, Q7)

remained de-energized, and consequently the irrigation pump (Q8) stayed off, confirming that the pump correctly reflects the OR condition of the three valves and does not activate in the absence of any dry-soil trigger (Fig. 9).

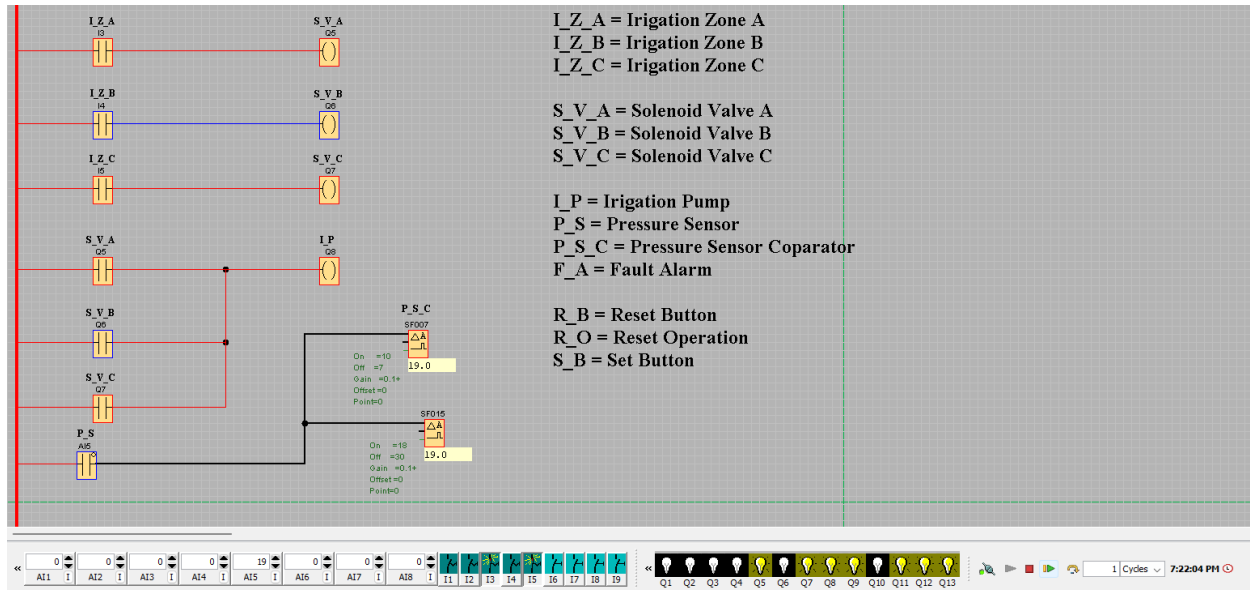


Figure 9: Ladder Diagram for irrigation zone operation

With the ambient light inputs for both zones (AI7, AI8) held above their respective thresholds, comparators SF010 and SF011 remained active, keeping both grow lights (Q12, Q13) off under sufficient natural light, confirming that the lighting subsystem does not needlessly activate when ambient light is adequate (Fig. 10).

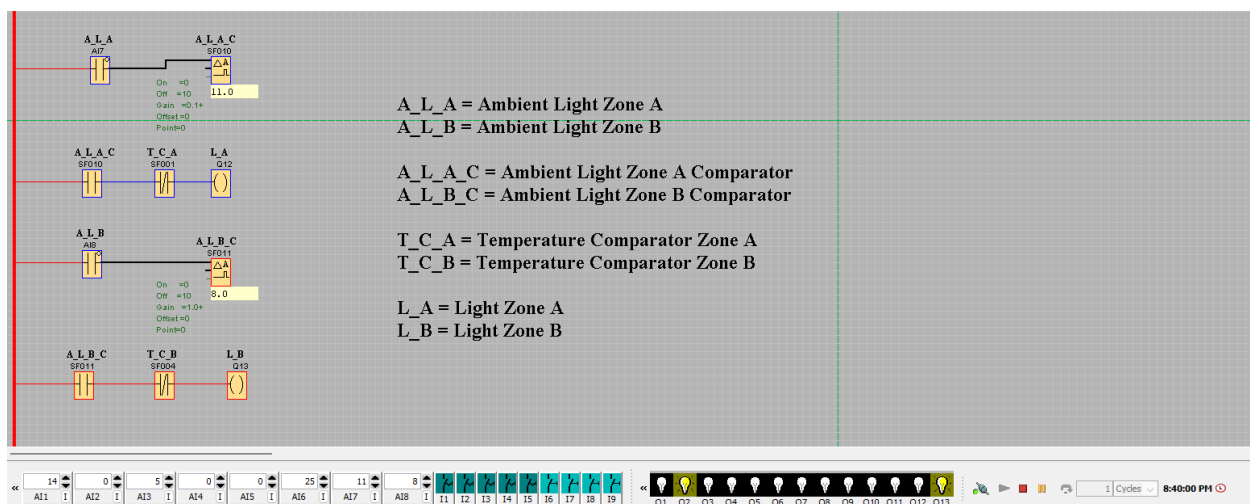


Figure 10: Ladder Diagram for ambient light zone

Over-Temperature Scenario: Raising the Zone A temperature input (AI1) above its threshold activated comparator SF001, energizing the circulation fan (Q1) for its timer duration and opening the vent motor (Q2), provided the rain sensor (I2) remained inactive. Lowering the temperature back below the threshold de-energized the vent motor, while the fan continued running until its timer elapsed, verifying the independent timing behaviour of the two actuators (Fig. 11).

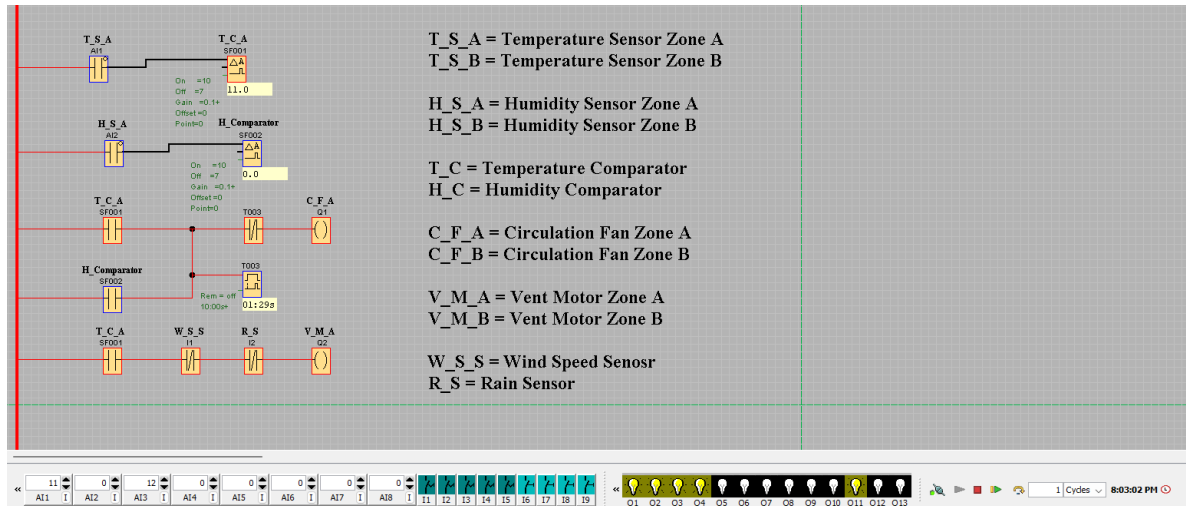


Figure 11: Ladder Diagram for over-temperature scenario of zone A

Repeating the test on the Zone B temperature input (AI3) produced identical behaviour through comparator SF004, activating the circulation fan (Q3) via timer T006 and opening the vent motor (Q4), again subject to the rain sensor (I2). This confirmed that Zone A and Zone B operate independently under identical over-temperature logic, with no cross-triggering between zones (Fig. 12).

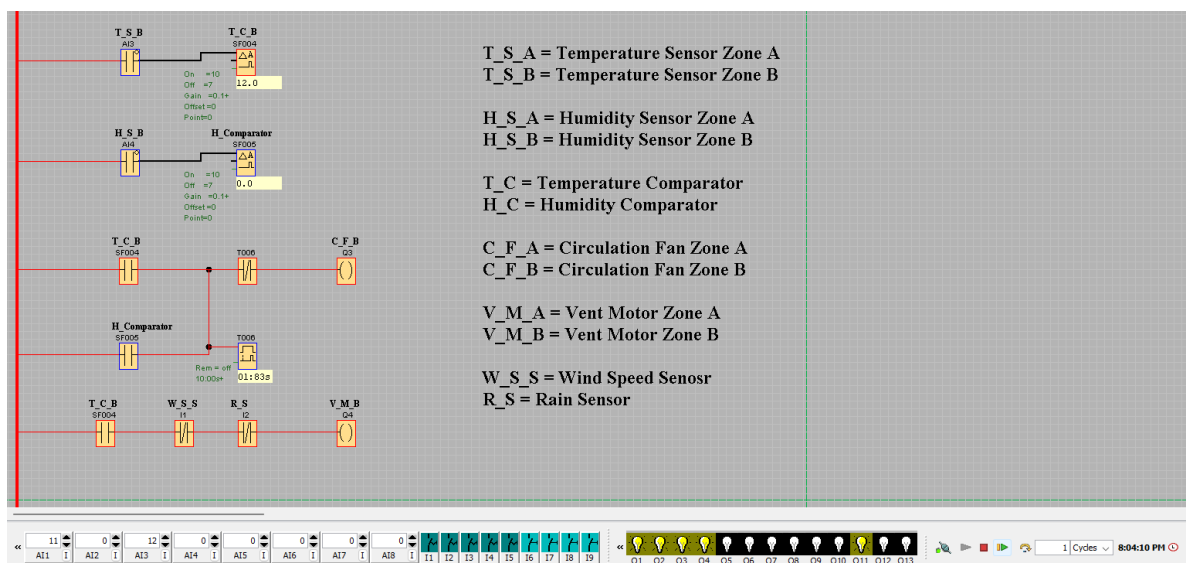


Figure 12: Ladder Diagram for over-temperature scenario of zone B

To verify temperature-based inhibition of the grow lighting subsystem, the Zone A ambient light input (AI7) was set low while the temperature input (AI1) was raised above threshold. The grow light (Q12) stayed off despite insufficient light, as SF001 opened the inhibiting contact. Lowering the temperature back below threshold, with light still low, restored the grow light, confirming that temperature inhibition takes priority as intended (Fig. 13).

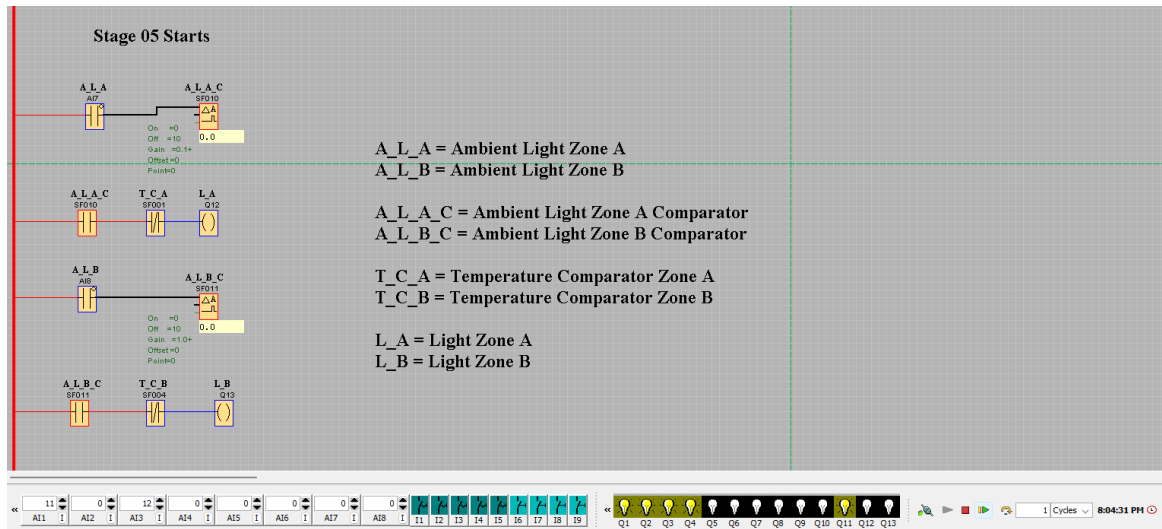


Figure 13: Ladder Diagram for over-temperature scenario of ambient light zone

Soil Dry Scenario: Setting the Zone A (I3) and Zone C (I5) soil moisture inputs to dry energized valves Q5 and Q7 and activated the pump (Q8), while Zone B remained unaffected. Restoring both to wet closed the valves and stopped the pump, confirming independent zone operation and correct OR-based pump logic (Fig. 14).

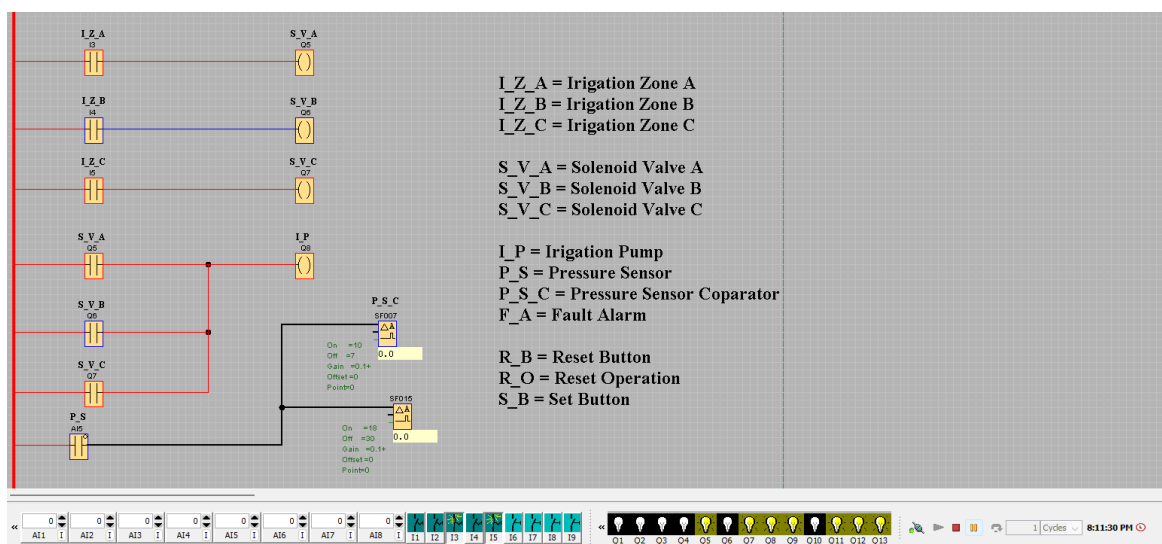


Figure 14: Ladder Diagram for soil dey scenario

Fertilizer Empty / Pump Fault Scenario: With the irrigation pump (Q8) active, lowering the fertilizer tank level input (AI6) below the threshold caused comparator SF009 to trigger, immediately closing the dosing valve (Q10) and arming the associated fault latch through the set button (I8). The alarm condition persisted even after the tank level was restored, and was cleared only after the reset button (I9) was pressed, while the irrigation valves and pump continued operating normally throughout the fault, confirming that the fault is correctly isolated to the fertilizer dosing subsystem (Fig. 15).

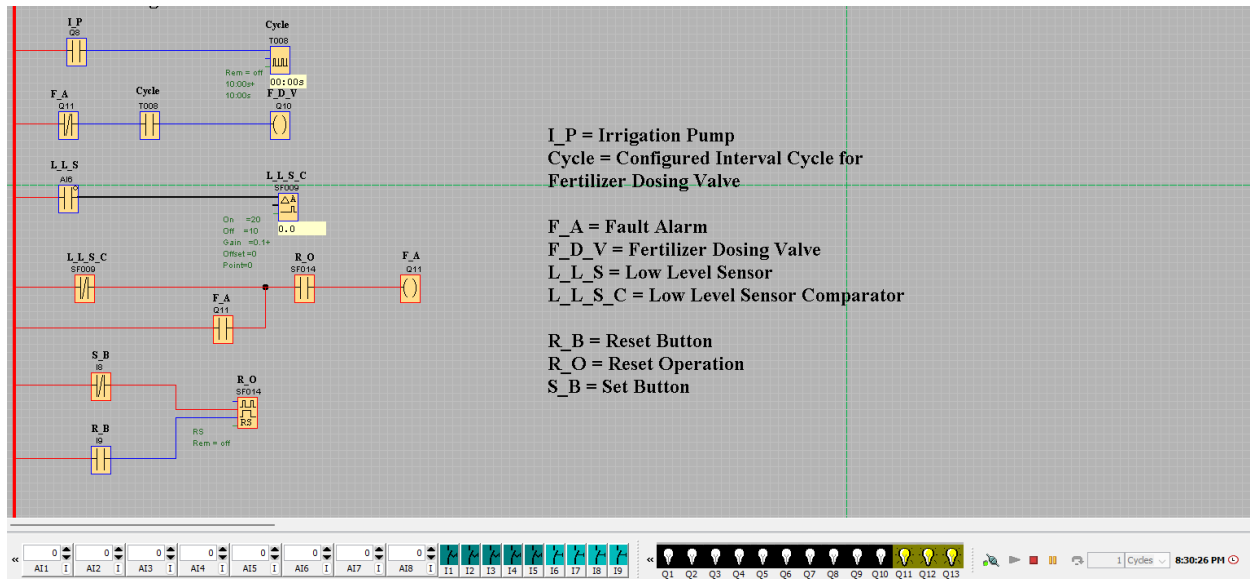


Figure 15: Ladder Diagram for Fertilizer Empty Scenario

To simulate a pump fault condition, the irrigation line pressure input (AI5) was raised above the high-pressure threshold while the pump (Q8) remained active, causing comparator SF015 to trigger and arming the corresponding fault latch through the set button (I6). The alarm condition remained active even after the pressure was brought back within the normal range, and was cleared only after the reset button (I7) was pressed, while the fertilizer dosing subsystem and grow lighting continued operating normally throughout the fault, confirming that the pump fault condition is correctly isolated to the irrigation pressure monitoring logic (Fig. 16).

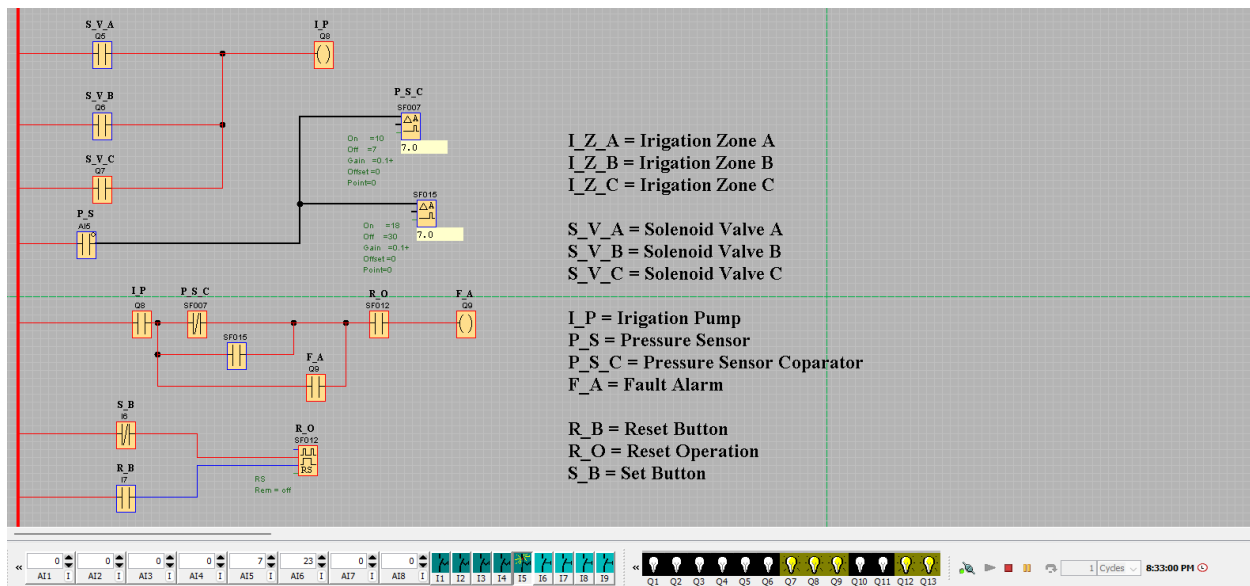


Figure 16: Ladder Diagram for Pump Fault Scenario

Impact of Precision Greenhouse Automation

Societal Impact: The automation of greenhouse operations directly addresses one of the most pressing societal challenges of our time — food security in the face of a growing population and shrinking arable land [4]. By replacing manual, labour-intensive monitoring of climate, irrigation, and fertilization with a PLC-based control system, the proposed design reduces dependency on constant human supervision, which is particularly valuable in regions like Bangladesh where seasonal labour shortages and unpredictable weather often disrupt traditional farming practices. The system's ability to operate continuously and respond to environmental changes in real time also improves consistency in crop yield, helping small and medium-scale farmers achieve more predictable outcomes regardless of external labour availability [7]. Furthermore, as automation becomes more embedded in agricultural practice, it creates a demand for a new category of technically skilled workers; operators and technicians capable of maintaining and troubleshooting such systems encouraging a gradual shift toward more technology-literate rural employment.

Environmental Impact: From an environmental standpoint, the zone-based irrigation strategy implemented in this system ensures that water is delivered only to sections of the greenhouse where soil moisture sensors detect an actual deficit, rather than applying water uniformly across the entire cultivation area [6]. This targeted approach significantly reduces water wastage compared to conventional irrigation methods. Similarly, the fertilizer dosing subsystem is designed to operate only when the irrigation pump is active, ensuring that fertilizer is always properly diluted before entering the soil, which minimizes the risk of chemical runoff and nutrient leaching into surrounding

ecosystems [5]. The use of timed operation for circulation fans and grow lights further contributes to energy conservation, as these actuators are not left running continuously but are instead activated only when climatic conditions genuinely require intervention. Collectively, these design choices reflect the broader goal of precision agriculture; achieving higher productivity while conserving the natural resources on which long-term agricultural sustainability depends [4].

Conclusion

This report presented the design, implementation, and simulation of an automated greenhouse control system using ladder logic in LOGO!Soft Comfort V7. The system integrates five interconnected stages; climate and humidity control, precision zone irrigation, fertilizer dosing, grow lighting control, and fault detection with recovery into a single coherent automation framework. Each stage was developed based on independent zone-wise logic, using analog comparators for threshold-based sensor evaluation, timers for controlled actuator duration, and latching circuits for persistent fault handling, closely reflecting the behaviour of real industrial PLC systems [1].

The system was validated through four simulation scenarios: normal operation, over-temperature response, soil dry condition, and fertilizer/pump fault handling. All four scenarios produced the expected behaviour, confirming that the climate, irrigation, and lighting subsystems responded correctly to their respective sensor inputs, while the fault subsystem correctly latched alarms and required manual reset before restoring normal operation. During development, a few implementation issues were encountered and resolved, particularly around correctly wiring analog comparator outputs and structuring hysteresis thresholds to avoid actuator chattering. The most significant learning challenge, however, was designing the fault latching logic using the Set-Reset (SR) mechanism understanding how to seal a fault condition so that it persists independently of the original trigger, and clears only upon deliberate operator reset, required a deeper appreciation of memory-based ladder logic compared to simple combinational rungs.

Overall, this assignment reinforced practical PLC programming skills, including analog signal conditioning, timer-based sequencing, and fault-tolerant system design, while also highlighting the broader relevance of automation in addressing real-world agricultural challenges. The exercise demonstrated that even a relatively compact PLC platform can effectively manage a moderately complex, multi-stage industrial process when the logic is carefully structured and validated through systematic simulation testing.

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